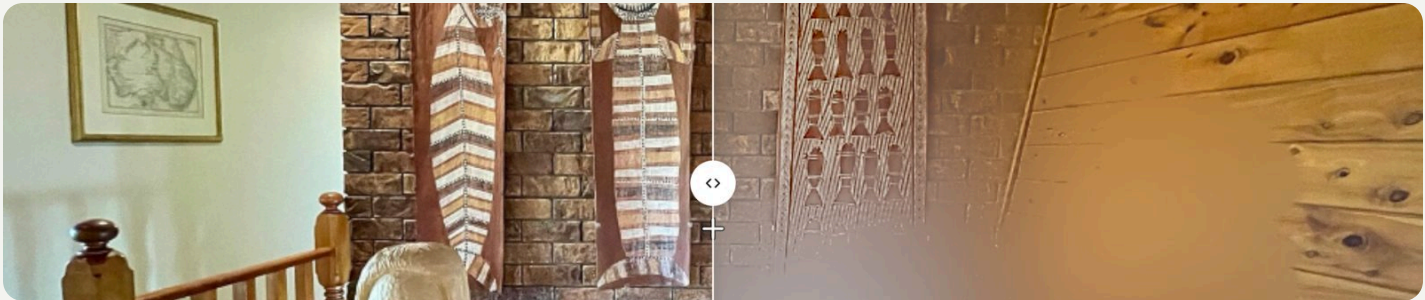


See the room and hear your voice the way someone with glaucoma and hearing loss does.

From their own visual field test and audiogram. Then plan together what to change at home and how to talk with each other.



Left: a hallway as you see it. Right: as “Meera” sees it from a real visual field test. Nothing goes black; detail fades and objects disappear into their surroundings.

The problem

Public images of glaucoma show a black tunnel. When 50 patients compared pictures with their own sight, none chose it (Crabb 2013). Hearing loss, the largest modifiable dementia risk factor, is invisible, and families become someone’s ears and eyes with no guidance.

Designed with dignity

Simulations alone can make people rate disabled people as less capable (Silverman 2015). Every screen pairs the simulation with what still works and how people adapt, uses real patients’ words, and the person co-owns the plan.

See

The camera through their field: Peli’s band-limited contrast model on the GPU, thresholds raised by $10^{-TD/10}$ from their total deviation. Years slider; evening and TV light with CIE glare; on a laptop it follows your eyes.

Hear

Their audiogram drives the Cambridge MSBG simulator; Whisper, behind a noise floor at the human hearing threshold, reports what they likely heard. No audiogram? A paired beep test on shared headphones gives one. Claude rewordings are kept only if they score better.

Walk • Sit • Guide

On-device depth finds step edges (0.56 false edges per stair-free room on held-out photos) and compares their contrast with what their vision needs there. Seating by better ear and clear field. One editable page for the family.

Does the hearing model match real people?

Clarity Prediction Challenge 2 evaluation set: 897 sentences, 15 listeners with hearing loss, predicting the share of words each person repeated correctly.

Method	RMSE	Correlation
Our proxy listener, Whisper small.en (calibrated)	26.9	0.74
Deployed configuration (Whisper base.en)	27.7	0.72
HASPI, standard intelligibility index (calibrated)	28.6	0.70
Best published system (trained on CPC2 data)	25.1	0.78

Honest limits. The vision view is exact only while looking at the fixation cross; the 24-2 test covers the central 24–30°. The hearing model is validated on CPC2 listeners, not on the profiled person. Dim light assumes lamp brightness; the in-app hearing test is about ±10 dB; printout reading is checked on synthetic printouts, and the person verifies every number. Not a diagnostic tool. **Next:** measure eye tracking across real webcams, test printout reading on real printouts, and sessions with older adults, an optometrist, an audiologist and an OT.